



Research Practice Presentation

Keyword-Auction Bidding Optimizer for Amazon Advertising

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Introduction

- **Background**

In order to make sellers advertise their products to the customer base, amazon provides a real-time bidding-based display advertising arrangement. RTB enables advertisers to buy individual ad impressions via an auction in real-time and facilitates the presentation and the bidding of individual impressions across multiple advertisers.

A Pay-per-click model is followed here. These are second-price auctions.

PPC advertising includes various formats (e.g., search, contextual, social) with a total investment of more than 200^[8] billion USD per year worldwide.



Introduction

- **Motivation:**

Amazon as an e-commerce platform provides a **keyword** based RTB facility for advertisers to reach out to their potential audience.

The advertisers' goal is to choose the **bid** and **daily budget** values for every campaign to maximize the revenue subject to a cumulative budget constraint over the campaign. This optimization problem is particularly challenging, as it includes many intricate subproblems such as,

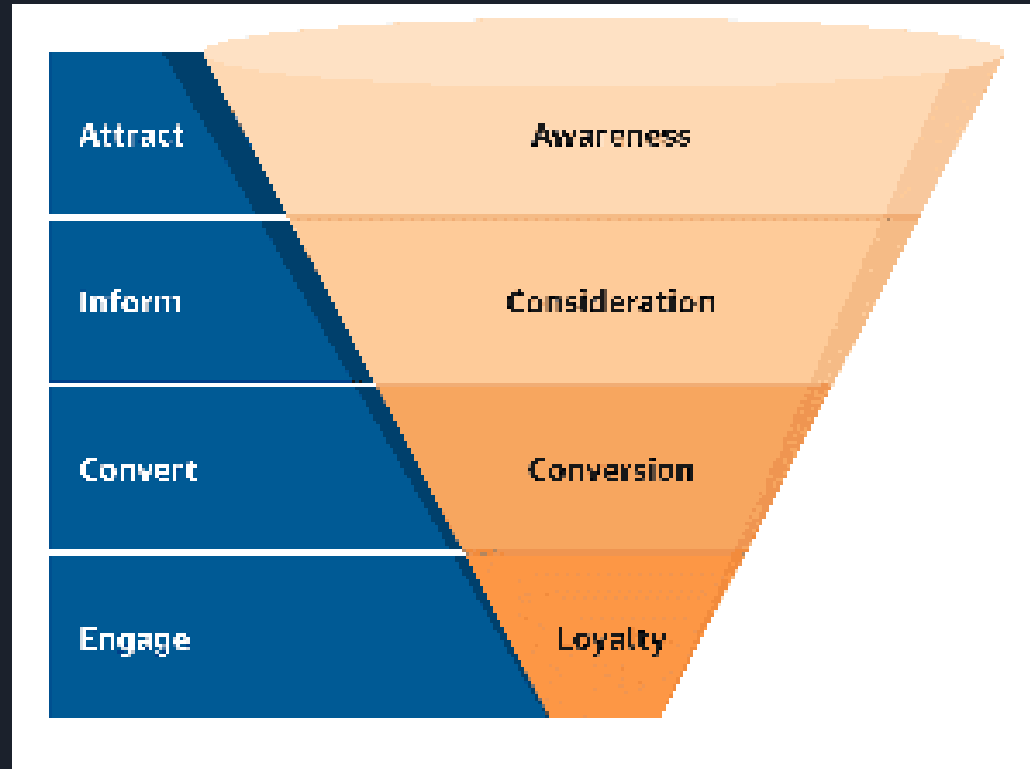
- finding the best bidding in auctions,
- estimating the value per click of the campaigns
- estimating conversions and profits for deducing an optimized ROAS.

Introduction

The Marketing Funnel:

The marketing funnel shape corresponds with the idea of journey that a shopper goes through, while making purchase from platform.

It also points about the, customer traffic that reduces through each subsequent stages.



Source: <https://advertising.amazon.com/library/guides/marketing-funnel>

Source: Knowledge Document - Advertising Metrics

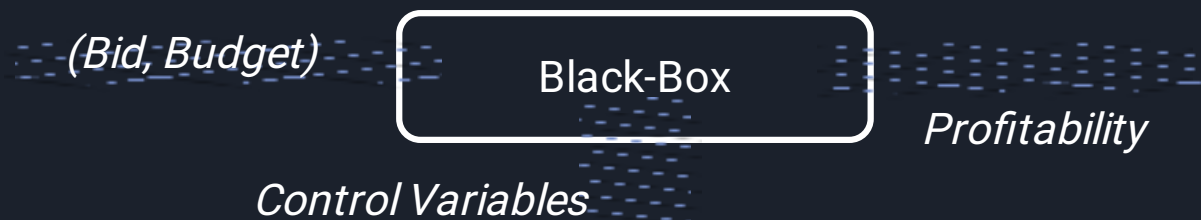
Introduction

- **Objective:**

The aim is to identify and explain the changes of inputs in a given advertising campaign on its output evaluative parameters (ROAS, Impressions, CTR).

This got further focused towards developing an prediction model system, that given a set of ad-campaign input specifics (particularly: *Bid, Budget*) can predict end-measure 'profitability' of the campaign in terms of multi-class manner.

Further expansion sees, a what if analysis inspired decision tree, that tells results of various input changes on to the profits.





Related Work

1. Cetină et al. [1] attributed various factors that influence the buying decision of customers on e-commerce portals. Eg: Demographical, Economical, Marketing factors, Web-experience, online perception to name a few. Bala et al. [2] tried to understand internet marketing & its techniques with the objective of how can internet marketing can be used more efficiently to induced more audience and new developments in the field of tech that help it.
2. D Gunawan et al. [3] discussed classifying web-documents using fast text based similarity to deduce similarity amongst documents, used cosine similarity based on frequency of words.

*Source: 1. Psychological and Social Factors that Influence Online Consumer Behavior by Iuliana Cetină
2. A Critical Review of Digital Marketing. International Journal of Management by Bala, M*



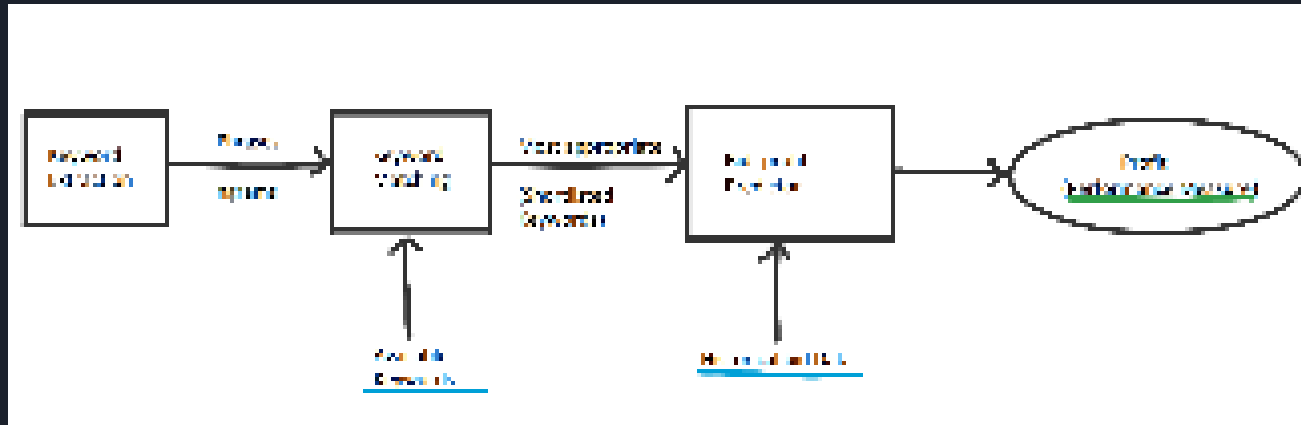
Related Work

3. Khandelwal et al.[4] considered various bidding strategies for Amazon-EC2 based spot instances. They considered several requirements such as hardware, latency, deadline and budget constraints. It highlights the objectives of each strategy and provides the suitability of each of the proposed bidding strategies based on the type of application.
4. Song et al.[6] gave a mechanism that depends on a repeated uniform price auction. It also included bidding adjustment model. They show that uniform price auction achieves optimal efficiency outperforming current amazon-spot market avg-revenue by 14%.
5. Shi et al.[7] attempted to predict CTR and avg CPC for keywords based upon their previous performance. For this, they used Google-Adwords provided data, which was at keyword-granularity. Employed Linear Regressor, Random Forest, Gradient Boosting techniques to predict the results, produced results with *mse* being in accepting bounds. They have used *MSE* as their criteria of evaluation.

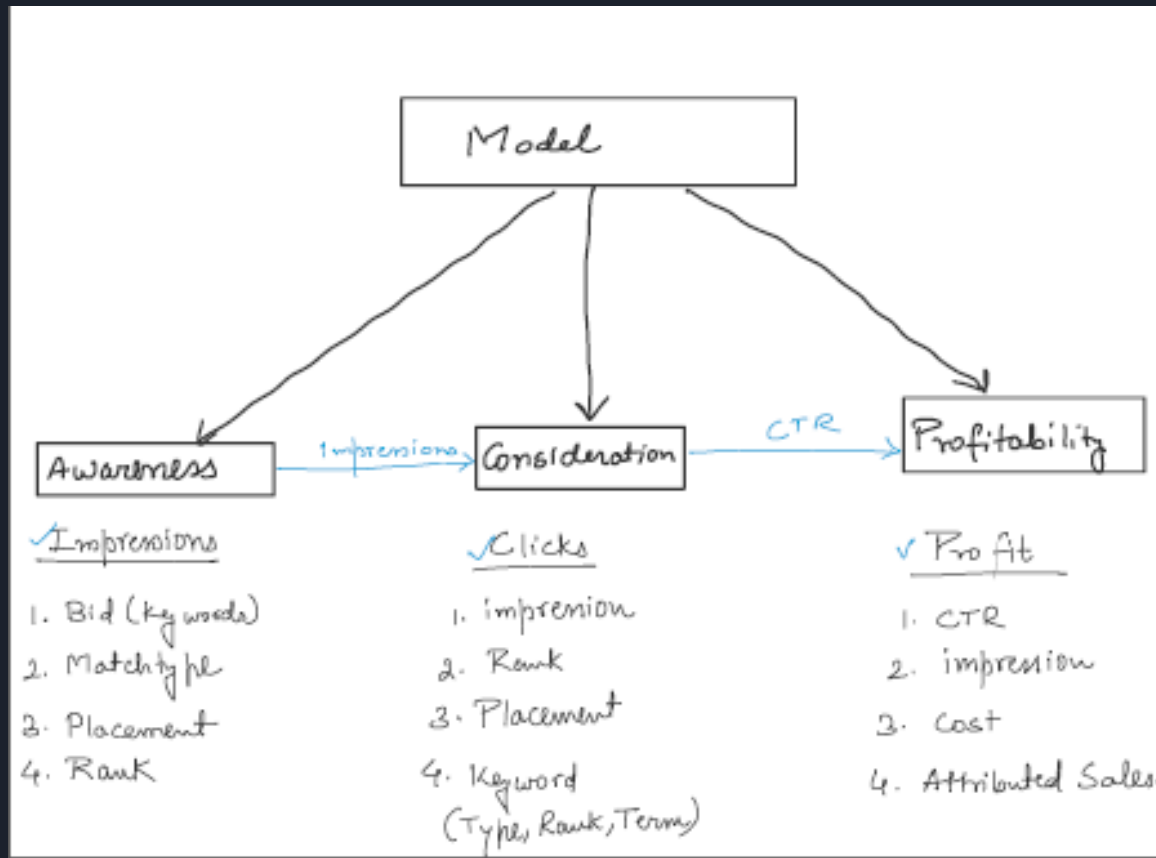
Proposed technique and Algorithm

During early phase of discussions, came up with a multi-module system keeping comprehensive algorithmic optimizer in focus. The proposed architecture consists of 3 modules:

- Keyword Extraction Module:
- Keyword Matching Module
- Bid-Strategy Optimizer



Later, we focused on the sub-problem in current phase, i.e., *given the specifics of a campaign (bids, budget) predict its profitability*. For this, we divided the problem statement into 3 parts: each dealing with one of the stage from advertisement funnel, which are Awareness phase, Consideration phase, Conversion phase.





Proposed technique and Algorithm

1. Impression Classification:

Since increased bid-amount will result in increased chances on getting into winning-pool, that implies more chances of ad-being shown (Impression). So increase in Impression results in inflated subsequent metrics as well, ultimately increased profits. This module takes bid, Match-type and SKU-rank as its input, and predicts Impressions as a multi-class. *Dtree*, *RF-Classifer* (with grid-search) and *multilayer-perceptron* based classifier were used.

Another module is tested with *RF-based regressor* to forecast impressions (*numerics*) based on bid, budgets as changing inputs and other control variables. [Link](#)

2. Click Classification:

Following in the similar intuition, the increase in Impression drives more clicks. To realise it, we used Campaign-Keyword level data while only considering keywords with 'Enabled' status. Among 4 classifier applied RF-based classifier gave better accuracy.

Similar module is tested to predict the values of Clicks, based on the inputs and predicted impressions (Impression') deduced from them. [Link](#)



Proposed technique and Algorithm (cont..)

3. Cost Prediction :

The predicted values of impressions, clicks is used solely to deduce output. Due to its performance, *RF-based regressor* is used here. Also, no control variable are used here.

4. Sales Prediction :

The predicted values of (impressions, clicks) is used solely to deduce output. Due to its performance, *RF-based regressor* is used. Also, no control variable are used here. It has been tested against actuals values of attributed_sales, to claim its accuracy.

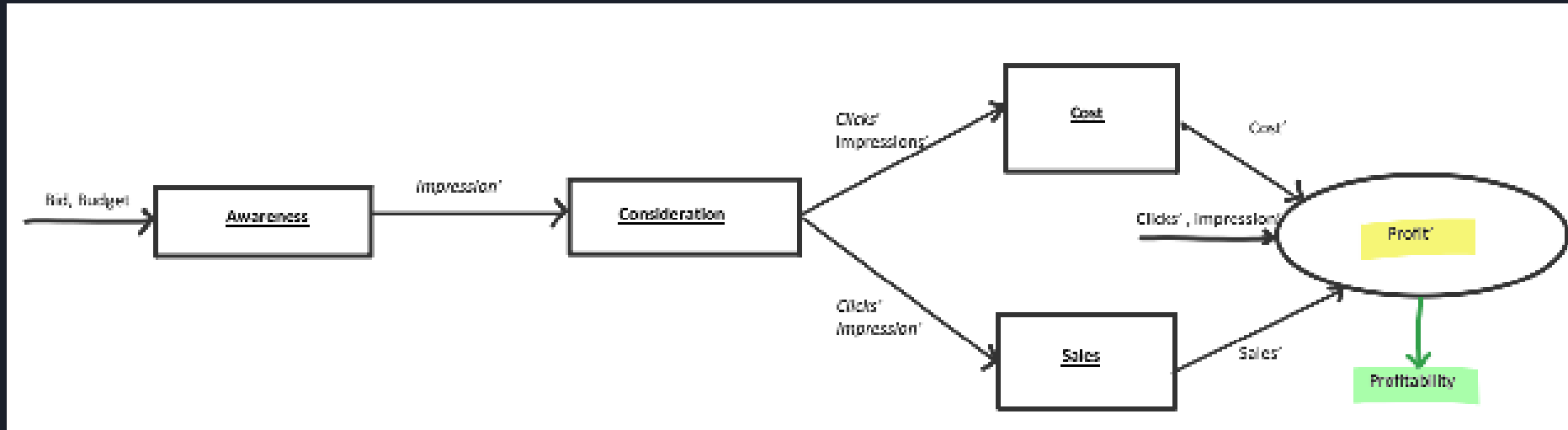
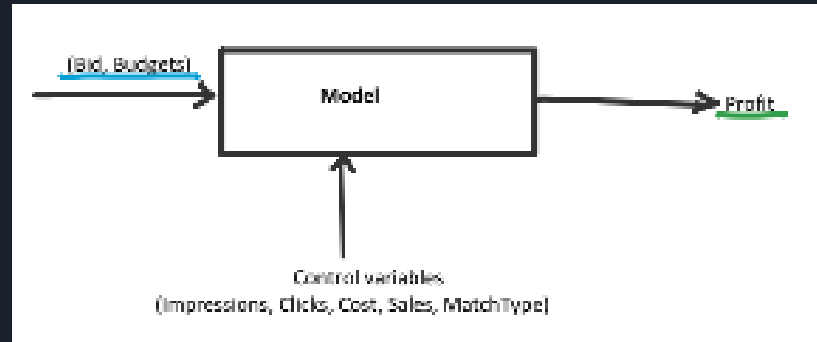
5. Profitability Prediction :

Estimated values of clicks, impressions, costs, sales are used to predict profits from a given input-set. The formula used:

Profit = (RPC-CPC) * impression * CTR /100

Profitability= (estimated-profits/ baselined profit – 1) *100

Proposed technique





Datasets used

- The dataset used for the Keyword extraction model and sentimental analysis is, Amazon-product review dataset^[1] from Kaggle. It contained, 34000 reviews for 42 different **asins** along with other rating, recommendations, product-related information.
- Amazon-mobile-electronics dataset^[2] was used from sentimental analysis using DistilBert. It contained 104974 reviews for products.
- The business has provided the historical data, which is hosted on a snowflake-dw. It contains data at various granularity: keyword level, ad-group level, campaigns, also data related to keyword searches and product-ranks included.

Source: <https://www.kaggle.com/datasets/datafiniti/consumer-reviews-of-amazon-products>
: https://www.tensorflow.org/datasets/overview/huggingface:amazon_us_reviews/

Datasets used

A brief on the tables provided by business:

1. **keywords_report** : Daily records of keywords within SP campaigns. Performance of keywords at campaign level granularity present.
2. **keywords_headline_report** : Daily records of keywords-campaign is present within SB campaigns. Keyword-adgroup-campaign level.
3. **asins_report** : Daily record of asins/products in SP campaigns. Keyword-product-campaign mapping.
4. **campaigns_report** : Daily reports at campaign level in SP campaigns. Campaign specifics and metrics available, but *campaign-bid mapping* is absent.
5. **campaigns_headline_report** : Daily reports at campaign level in SB campaigns, mapping of campaign-costs-sales-budgets is available.
6. **search_data_archive** : Daily scraped data of selected search terms, contains ranking of products, specifics of the competitor and product.
7. **search_data_info** : List of search terms those are scraped daily by business, also a mapping of keyword-category is available.
8. **keyword_classification** : contains keyword classification for a given keyword, keywords are internally classified by business among: client_branded, competitor_branded or generic.

Source: [Document containing information about data and comments on columns of these tables](#)

Experiments and Results:

- Evaluation and Results:

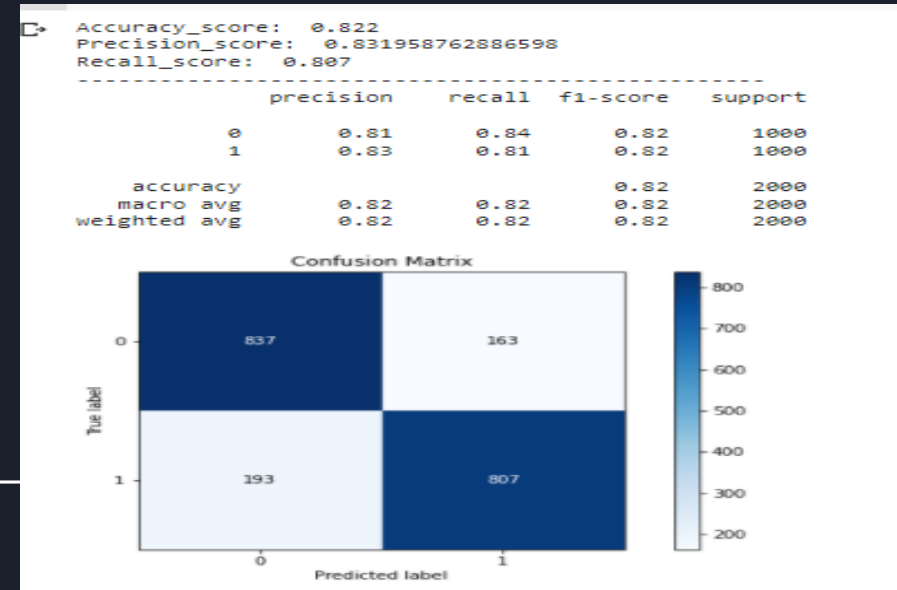
1. With TF-IDF based Key-Word extraction, lessons learnt were: **negative word** be omitted, users **don't search** what they have to **say** about the product but **what they want**.
2. The Sentiment-Classification model using the Pretrained DistilBERT model gave an accuracy of 74.7%.
3. The BOW-based RF model when ran over Electronic data, gave an accuracy of 82.2%.

```
Amazon_Product_Review_using_BERT.ipynb ☆
File Edit View Insert Runtime Tools Help All changes saved

+ Code + Text

acc_score = accuracy_score(label,validation_labels)
pre_score = precision_score(label,validation_labels)
rec_score = recall_score(label,validation_labels)
print('Accuracy_score: ',acc_score)
print('Precision_score: ',pre_score)
print('Recall_score: ',rec_score)
print("-"*50)
cr = classification_report(label,validation_labels)
print(cr)

Accuracy_score: 0.747
Precision_score: 0.9893617021276596
Recall_score: 0.7522750252780587
-----
```



Experiments and Results (cont.):

4. The impression classification model was implemented with Dtree, RF, NN-sgd 4-class classification. Inputs(keyword_bid, search_type, search_rank)

Results:

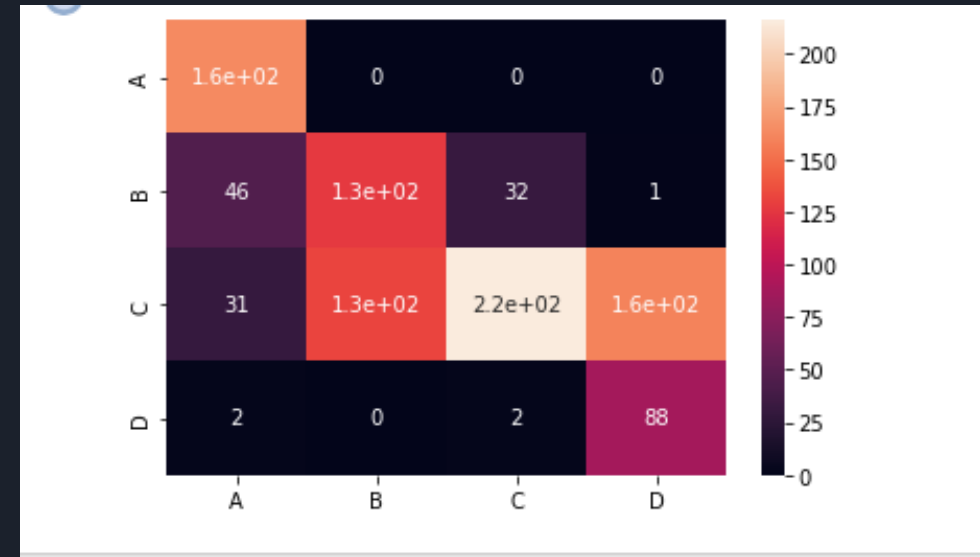
Models	Accuracy
Dtree	54.6
RF-classifier	57.6
NN (relu,adam)	59.4

```
Accuracy of NNClassifier
precision recall
A 1.00 0.67
B 0.00 0.00
C 0.40 0.80
D 0.00 0.35

accuracy
macro avg 0.74 0.68
weighted avg 0.74 0.68

Confusion Matrix
[[161 46 11 2]
 [ 0 127 112 0]
 [ 0 32 210 2]
 [ 0 1 160 88]]

Accuracy
0.804
```



Experiments and Results (cont.):

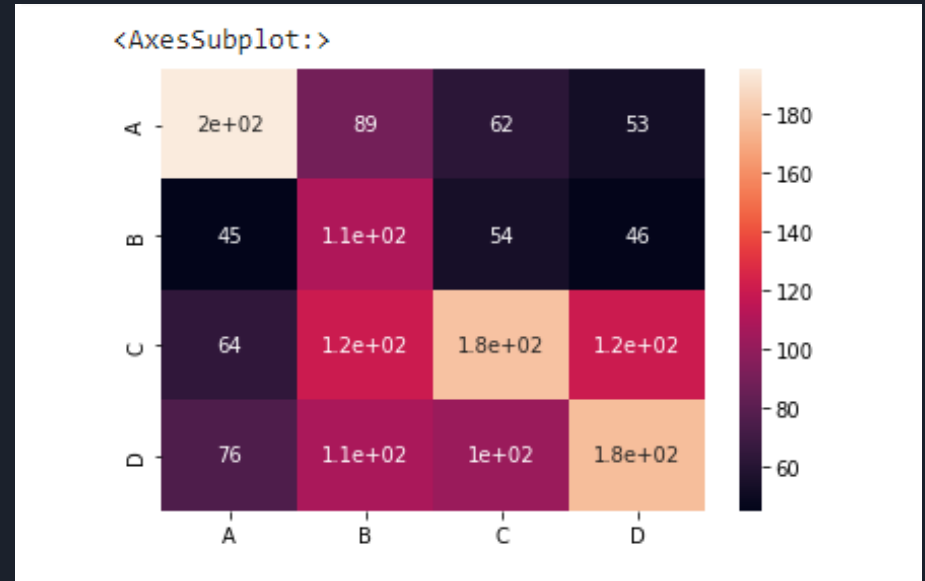
- The CTR classification model was implemented with Dtree, RF, NN-sgd
4-class classification. Inputs(Match_score, keyword_bid, impressions, rank)

Results:

Models	Accuracy
Dtree	40
RF-classifier	45.85
NN (relu,adam)	41.2

```
[ ] print(grid_search.best_score_)
    rf_best = grid_search.best_estimator_
    print(rf_best)

0.45859375
RandomForestClassifier(max_depth=10, min_samples_leaf=10, n_estimators=150,
                        n_jobs=-1, random_state=42)
```



Experiments and Results (cont.):

5. The RF-based regressor was used to predict, Sales=(impressions, Clicks). Acc: 40.94%
6. Similarly, RF-based regressor was used to predict, cost=(impressions, clicks). Acc: 99.9%

```
#####  
  
# Calculate mean absolute percentage error (MAPE)  
mape = 100 * (errors / y_test)  
# Calculate and display accuracy  
accuracy = 100 - np.mean(mape)  
print('Accuracy:', round(accuracy, 2), '%')
```

Mean Absolute Error: 61.53 degrees.
Accuracy: 40.94 %.

```
loss:  
0.014017
```

```
In [14]: keywords_names_of_interest()
```

```
Out[14]:
```

	CAMPAIGN_ID	IMPRESSIONS	COST	CLICKS	ATtributionValue
CAMPAIGN_ID	1000000	0.000000	0.000000	0.000000	0.000000
IMPRESSIONS	0.000000	1.000000	0.000000	0.000000	0.000000
COST	0.000000	0.000000	1.000000	0.000000	0.000000
CLICKS	0.000000	0.000000	0.000000	1.000000	0.000000
ATtributionValue	0.000000	0.000000	0.000000	0.000000	1.000000

750 rows x 2 columns

```
In [15]: # Calculate the absolute errors  
errors = abs(predictions - y_test)  
  
# Get Mean Squared Error  
print('Mean Absolute Error:', round(np.mean(errors), 2))
```

```
#####  
  
# Calculate mean absolute percentage error (MAPE)  
mape = 100 * (errors / y_test)  
# Calculate and display accuracy  
accuracy = 100 - np.mean(mape)  
print('Accuracy:', round(accuracy, 2), '%')
```

Mean Absolute Error: 8.04
Accuracy: 99.95 %.

Experiments and Results (cont.):

7. To predict Impression as intermediate of (bid,budget) inputs.
Impression=(*BID, CAMPAIGN_BUDGET, PROFIT, CTR,COST, SALES*). Acc: 85.51%
8. Similarly, click=(*BID, CAMPAIGN_BUDGET, Impression, PROFIT, CTR,COST, SALES*)
. Acc: 99.9%

```
# # Calculate the absolute errors
# errors = abs(predictions - y_test)

# Calculate mean absolute percentage error (MAPE)
mape = 100 * (errors / Y)
# Calculate and display accuracy
accuracy = 100 - np.mean(mape)
print('Accuracy:', round(accuracy, 2), '%.')
```

Mean Absolute Error: 188.47 degrees.
Accuracy: 85.51 %.

```
# errors = abs(predictions - y_test)

# Calculate mean absolute percentage error (MAPE)
mape = 100 * (errors / y_test)
# Calculate and display accuracy
accuracy = 100 - np.mean(mape)
print('Accuracy:', round(accuracy, 2), '%.')
```

Mean Absolute Error: 0.84 degrees.
Accuracy: 81.06 %.

Error Analysis:

1. For regressor prediction, we have used `mean_absolute_percentage_error`.
$$\text{MAPE} = (\text{predicted_value} - \text{actual_value}) * 100 / \text{actual_value}$$

By this, we make sure are able to get a reative scale for errors. This also, protects against large-delta that is originated from of difference relatively large values.

Also, it awards the case, where error is small, relative to actual-values: (0.5/99) case.
2. Earlier, we got Clasifcation_accuracy for Impressions as low as 0.37,0.44(best). Upon checking, came to know its was due to date being skewed towards A-class(max-impressions). Equally sampled records, resulted in 0.57(best) accuracy.
3. Earlier, we got Accuracy as 11% for Impression-predictor, because we were taking all possible (campaign_budgets, bids). Then we did a filter by only 1 campaign_id to get-uniform trend. Improved to 85.5%

Out[38]:

	CAMPAIGN_ID	KEYWORD_ID	COUNT(*)
0	144352058263994725	144276687421074460	672
1	144221050922434038	144364899479084370	672
2	144218402871724521	144229367185785810	672



Work Update

☐ **Work Done:**

- Introductory keyword extraction module using TF-IDF.
- Similarity calculation module to calculate similarity based on Frequency of texts in document and TF-IDF induced document-vectors.
- Sentimental analysis of given product using DistilBERT & BOW based approach.
- Correlation calculations backed with knowledge documents on data provided by business.
- Output metrics was formulated and programmed at shared repository.
- Impression and CTRs multi-class classification has been attempted.
- A model to predict profitability based on the campaign inputs is constructed. It contains various intermediate states (predictions of impressions, clicks, cost, sales) that in turn calculates profits.

☐ **Work Remaining:**

- Custom Decision tree initialization, to display Profitability metric in terms of inputs.
- All models to be integrated in pipelined fashion, with the aim of 1 click execution. (using scikit-learn pipeline module)



Conclusion and Future work:

- An initial attempt to solve the profitability prediction problem is made. Accuracy for some intermediaries has been acceptable, while others has to be verified from business and subject to internal discussion.
 - We can explore the possibility of introducing a UI based solution, which shall be capable to predict the profits-class based on input(bis, budgets) provided for the campaigns whose historical data is already available for training.
-



References

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4. Mazzucco, Michele & Dumas, Marlon. (2011). Achieving Performance and Availability Guarantees with Spot Instances. 296 - 303. 10.1109/HPCC.2011.46.
5. K. Song, Y. Yao, and L. Golubchik, "Improving the Revenue, Efficiency and Reliability in Data Center Spot Market: A Truthful Mechanism," 2013 IEEE 21st International Symposium on Modelling, Analysis and Simulation of Computer and Telecommunication Systems, Aug. 2013.
6. Lihui Shi, Bo Li, Predict the click-through rate and average cost per click for keywords using machine learning methodologies, *Proceedings of the International Conference on Industrial Engineering and Operations Management* Detroit, Michigan, USA, 2016
7. Alessandro Nuara, Francesco Trovò, Nicola Gatti, Marcello Restelli, Online joint bid/daily budget optimization of Internet advertising campaigns, *Artificial Intelligence*, Volume 305, 2022, 103663, ISSN 0004-3702

Thank You

